THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Add 14 and its reverse 41. The sum is: 1

(a) 55 **(b)** 77 **(c)** 66 **(d)** 44

2. Kaprekar step: for 3752, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 5166 **(b)** 5265 **(c)** 5184 **(d)** 5175

3. In the hailstone (Collatz) sequence, the number after 18 is: 1

(a) 36 **(b)** 10 **(c)** 9 **(d)** 55

4. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 57: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

5. The digit sum of 4567 is: 1

(a) 18 **(b)** 31 **(c)** 29 **(d)** 22

6. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 13 and 18, the number above them is: 1

(a) 31 **(b)** 234 **(c)** 41 **(d)** 5

7. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 4, 6, 3. The top number is: 1

(a) 13 **(b)** 25 **(c)** 26 **(d)** 19

8. Add 25 and its reverse 52. The sum is: 1

(a) 99 **(b)** 66 **(c)** 88 **(d)** 77

9. Kaprekar step: for 8261, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 7362 **(b)** 7344 **(c)** 7353 **(d)** 7443

10. In the hailstone (Collatz) sequence, the number after 11 is: 1

(a) 5 **(b)** 34 **(c)** 35 **(d)** 33

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Apply the reverse-and-add process on 47 until a palindrome appears. 2

12. Do one Kaprekar step on 2961. 2

13. Write the hailstone (Collatz) sequence for 6. 2

14. Apply the reverse-and-add process on 58 until a palindrome appears. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the Kaprekar routine on 3524 reaches 6174. 3

16. Find the number of steps taken by the Collatz sequence starting at 13 to reach 1 and the largest number in it. 3

17. A number is formed by writing the digit 7 exactly 5 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

OR

Show that the Kaprekar routine on 4821 reaches 6174. 3

18. Find the number of steps taken by the Collatz sequence starting at 14 to reach 1 and the largest number in it. 3

OR

A number is formed by writing the digit 8 exactly 6 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take the digits of 3141. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 121 and 232 (both palindromes) gives 353. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 29. 5

20. Take the digits of 2718. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 343 and 454 (both palindromes) gives 797. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 38. 5

END OF QUESTION PAPER — Total: 40 Marks

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